

Human Monkeypox Virus Infection is a Potential Threat: How Ready are Nigerian Ophthalmologists?

Monsudi KF,¹ Osayande-Osawe O,² Mustapha T,³ Salahu IA⁴

ABSTRACT

Mpox is a zoonotic disease that is caused by the monkeypox virus. The disease originated in Africa (Democratic Republic of Congo), and it spread to other countries. Mpox was designated a public health emergency of international concern (PHEIC) in 2022 and 2024. Its modes of transmission are from animal to human and from human to human. The clinical presentation of mpox is similar to smallpox, cowpox, and chickenpox virus infection. The clinical presentation includes systemic fever, headache, myalgia, back pain, lymphadenopathy, and maculopapular rashes in the face, palm, and sole. The ocular manifestations of mpox include photophobia, redness, lid edema, blepharitis, sub-conjunctiva nodes, keratitis, cornea perforation, and blindness. The aim of this article is to create awareness on ocular features, management and prevention among ophthalmologist in Nigeria. The eye care workers especially ophthalmologists are prone to mpox during the management of an infective person. There is no available standard care treatment for mpox to date except for supportive care and symptomatic management. The use of personal protective equipment, good hand hygiene, sterilization/cleaning of instruments used in the eye clinic, isolating the suspected or confirmed cases; standard hospital waste disposal, and creating awareness of mpox are various ways of preventing strategies of this infection in the eye care service delivery. Ophthalmologists should be aware of ocular manifestation, prevention strategies, and management of this disease.

Keywords: Monkeypox, Zoonotic disease, Ophthalmologist, Smallpox vaccine, Nigeria

INTRODUCTION

Mpox is a zoonotic disease first discovered in 1958 in a monkey. The human infection was first reported in the Democratic Republic of Congo (DRC) in 1970.¹ The disease is caused by the monkeypox virus of the Orthopoxvirus genus (family Poxviridae) in the same group as smallpox.^{2,3} Mpox was designated a public health emergency of international concern (PHEIC) in 2022 and 2024.⁴ The human-to-human transmission outbreak was first reported in 1996–1997 in the Democratic Republic of Congo.² The first case of the disease reported in developed countries was in the USA in 2003.⁵ However, the first case in Nigeria

was reported in 1971.^{6,7} The disease's incubation period ranges from 5 to 21 days⁸ and it is usually a self-limited disease with symptoms lasting from two to four weeks.⁹ There are two strains of the monkeypox virus: Clades I and II formerly referred to as the Central African and West African strains respectively.¹⁰ The clade I cause a mild form of the disease while, the clade II results in a severe disease.¹⁰

The aim of this article is to create awareness on ocular features, management and prevention of mpox among ophthalmologists in Nigeria.

MODE OF TRANSMISSION

The modes of transmission of the disease are animal to human (through eating and contact); human to human through contact with body fluid, skin lesions, respiratory droplets, blood products; and fomites. Transmission has also been reported through the birth canal and via the placenta (congenital mpox).¹¹

Department of Ophthalmology,¹ Federal Teaching Hospital Birnin Kebbi, Kebbi State. Department of Ophthalmology,² Federal Teaching Hospital Lokoja, Kogi State. Department of Ophthalmology,³ Modibbo Adama University Teaching Hospital, Yola, Adamawa State. Department of Ophthalmology,⁴ Federal Medical Centre, Abeokuta, Ogun State.

*Corresponding Author: kfmshood@yahoo.com

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CLINICAL PRESENTATION OF MPOX

Clinical presentations of the mpox include systemic symptoms (flu-like symptoms; fever, myalgia, lethargy, headache, lymphadenopathy, widespread maculopapular rashes on the face, palm, and sole). The ocular presentation includes ocular pain, foreign body sensation, redness, photophobia, tearing, lid swelling, blepharitis, conjunctivitis, subconjunctival node, keratitis, corneal ulceration, corneal perforation, and blindness.^{12, 13} Eyelid and conjunctival involvement are the common ocular manifestations of mpox.¹³ Conjunctivitis in mpox has been reported to be predictive of systemic involvement.¹⁴

The eye care workers, especially ophthalmologists, are prone to mpox because they may be the first person to see this patient. The spread of the infection to other patients is possible through non-proper sterilization and cleaning of instruments [slit lamp, IOP pressure check (Goldmann, Perkins, Schiotz, and iCare tonometer)] used between the patients in the eye clinic.

Ocular complications of mpox include corneal scars, perforations, and blindness. These complications occur commonly in unvaccinated populations with the smallpox vaccine.¹⁵

DIAGNOSIS OF MPOX

The diagnosis of mpox is not clinical though clinical findings are suggestive. Cases are only confirmed following investigations using a real time or convectional polymerase chain reaction (PCR) and virus isolation from cell culture of conjunctiva swabs, eyelids and skin lesions.

Suitable sample for mpox are surface and/or skin materials such as exudate swabs, lesion crusts. The swab should be rubbed vigorously against the lesion to obtain sufficient viral DNA.¹⁶ The swap samples should be taken from more than one location and different

lesion, so as to prevent false results (false positive or false negative).¹⁶

TREATMENT OF MPOX

There is no available standard care for the treatment of mpox to date except for supportive care and symptomatic management.^{13,17,18} The treatment of ocular manifestations of mpox includes the use of anti-inflammatory (NSAID), prophylactic topical antibiotics, topical steroids, and antiviral drugs. Oral tecovirimat, an antiviral drug that works by inhibiting the viral envelope protein VP37 at the dose of 600 mg 12 hourly for 2 weeks, is widely accepted as effective in the treatment of ocular manifestations of mpox. Other antiviral drugs, such as trifluridine and cidofovir, have also been used but are not as effective as tecovirimat. Vitamin A drugs are also being used in the malnutrition patient.¹²

DIFFERENTIAL DIAGNOSIS OF MPOX

The differential diagnosis of mpox virus infection includes smallpox virus, herpes simplex virus, varicella-zoster virus, molluscum contagiosum infection¹², measles, scabies, syphilis, bacterial skin infection, and medical drug skin reaction.¹¹

The mpox can be transmitted through close contact with infected persons, contaminated objects, droplet transmission, and the potential for transmission through ocular secretions, hence ophthalmologists must learn about the prevention of nosocomial mpox.^{20, 21}

RECOMMENDATIONS FOR THE PREVENTION OF MPOX IN OPHTHALMOLOGY CLINICS

Ophthalmologists should have a high index of suspicion on any patient with fever and maculopapular rashes in the face, palm, and foot. They should ensure strict adherence to standard contact precautions and observe good hand hygiene practices (before and after each procedure/patient, in the eye clinic). There

is also the need for effective barrier nursing, the use of adequate personal protective equipment (glove, face mask, gown, and protective goggles) and ensuring proper treatment of any instrument used in the eye clinic, such as the Perkins, Goldmann tonometer, tonopen, iCare, and slit lamp, between patients in according to standard practice.^{12, 22}

Others preventive measure includes; the isolation of the suspicious or confirmed case in an isolated room/ward, report a suspected case to the hospital management, infection desk surveillance and notification officer. Contact tracing of all individual that directly or indirectly has interactions/contact with the confirmed case. Raising awareness and community mobilization, on preventive measures for mpox through hospital presentations, community meetings, and social media. Furthermore, proper and standard hospital waste disposal management and encourage acceptability of the smallpox vaccine by all eye care workers. The smallpox vaccine JYNNEOS (MVA-BN; Imvamune or Imvanex) and ACAM2000 are effective in preventing monkeypox disease.²² The JYNNEOS vaccine usage in Nigeria was approved by National Agency for Food Drug Administration and Control (NAFDAC) in 2023 subsequently available in the accredited treatment center in the country.²³

CONCLUSION

Mpox is a contagious disease that can make index presentation at the eye clinic hence; Ophthalmologists need to be aware of these ocular manifestations, prevention, and management strategies of the disease.

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